WORK SHEET - 28 **PROBABILITY**

Q.1 (D)

$P(A) = \frac{3}{4}$; $P(B) = \frac{2}{3}$

$P(A \cap B) = p$

As $P(A \cap B) \le P(B)$ and $P(A \cap B) \le P(A)$

$\therefore\ P(A \cap B) \le \frac{2}{3}$ ....(1)

Also $P(A \cup B) = \frac{3}{4} + \frac{2}{3} - p \le 1$

$\therefore\ p \ge \frac{5}{12}$ ....(2)

From (1) and (2) $p \in \left[\frac{5}{12}, \frac{2}{3}\right]$. **Ans**

Q.2 (A)

A need 2 more and B need 3 more points to win

$P\,(A \text{ wins}) = P(SS \text{ or } \underbrace{SF}\,S \text{ or } \underbrace{FFS}\,S) = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} = \frac{4+4+3}{16} = \frac{11}{16}$ **Ans.**

Q.3 (A)

For the required ordered pair (a, b) the number of values of 'b' corresponding to a particular value of 'a' is given by the following table

The value of 'a'	1	2	3	4	5	6	7	8
No. of values of 'b'	1	2	3	4	5	6	7	8

$\therefore$ The number of ordered pair (a, b), $n(A) = 1 + 2 + 3 + ... + 8$

$$n(A) = \frac{8 \cdot 9}{2} = 36$$

and total number of ways of selecting a pair from set P, $n(S) = 9 \times 223$

$\therefore$ Required probability $= \frac{n(A)}{n(S)} = \frac{36}{9 \times 223} = \frac{4}{223}$ **Ans.**

Q.4 (B)

For skew- symmetric matrix, $a_{ij} = -a_{ji}\ \forall\ i \ne j$ and $a_{ij} = 0\ \forall\ i = j$,

Now probability that a_{11} is $0 = \frac{1}{9}$, probability that $a_{21} = -a_{12} = \frac{9}{9} \times \frac{1}{9}$ is $\frac{1}{9}$, probability that a_{22} is $0 = \frac{1}{9}$.

probability that $a_{31} = -a_{13} = \frac{9}{9} \times \frac{1}{9}$ is $\frac{1}{9}$, probability that $a_{23} = -a_{32}$ is $\frac{1}{9}$, probability that $a_{33} = 0$ is $\frac{1}{9}$

$\therefore$ Required probability $= \left(\dfrac{1}{9}\right)^6$

Alternatively :

$n(S) = \underbrace{9 \times 9 \times \ldots\ldots \times 9}_{9\text{ times}} = 9^9$

Number of favourable cases $= (1 \times 9 \times 9 \times 1 \times 1 \times 9 \times 1 \times 1 \times 1) = 9^3$

Hence required probability $= \dfrac{9^3}{9^9} = \dfrac{1}{9^6}$ **Ans.**

Q.5 (D)

$p(0) = \dfrac{1}{3}$; p (all other faces) $= \dfrac{1}{6}$ each

E: sum vanishes

$p(E) = p\big((0,0) \text{ or } (1,-1) \text{ or } (i,-i)\big) = \dfrac{1}{3}\cdot\dfrac{1}{3} + 2\cdot\dfrac{1}{6}\cdot\dfrac{1}{6} + 2\cdot\dfrac{1}{6}\cdot\dfrac{1}{6} = \dfrac{4}{36} + \dfrac{2}{36} + \dfrac{2}{36} = \dfrac{8}{36}$ **Ans.**

Q.6 (B)

P (Exactly 5 games when A wins 4) $= P\left(\overbrace{\text{AAAB}}\ \underbrace{\text{A}}\right) = \dfrac{4}{16}\cdot\dfrac{1}{2} = \dfrac{1}{8}$

P (Exactly 5 games when B wins 4) $=$ P (BBBAB) $= \dfrac{4}{16}\cdot\dfrac{1}{2} = \dfrac{1}{8}$.

$\therefore$ required probability $= \dfrac{1}{8} + \dfrac{1}{8} = \dfrac{1}{4}$ **Ans.**

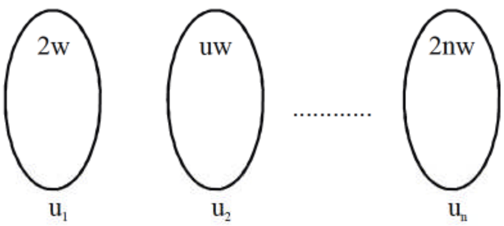
Q.7 (D)

If the first term, it can be one of 2, 3, 4 or 5 and there are 4 ! ways to arrange the other 4 digits, so there are $4 \cdot 4!$ elements in S.

If we want the second term to be 2, then there are 3 ways to choose a first digit, and 3 ! ways to arrange the other 3 digits, so there are $3 \cdot 3!$ element in this case.

The final probability is $\dfrac{3\cdot 3!}{4\cdot 4!} = \dfrac{3}{16}$. **Ans.**

Q.8 (A)



Total number of balls in each urn $= 2n + 1$

Let selecting i^{th} ball is $P(i) = k(i^2 + 2)$ k is proportionality constant

then $P(1) + P(2) + \ldots\ldots P(n) = 1$

$$\therefore\ k = \frac{1}{\left[\frac{n(n+1)(2n+1)}{6} + 2n\right]}$$

and $$P(A) = k(1^2+2)\cdot\frac{2}{2n+1} + k(2^2+2)\cdot\frac{4}{2n+1} + k(3^2+2)\cdot\frac{6}{2n+1} + \ldots\ldots k(n^2+2)\cdot\frac{2n}{2n+1}$$

$$= \frac{k}{2n+1}\cdot\sum_{r=1}^{n}(r^2+2)\cdot 2r$$

$$\therefore\ P(A) = \frac{\frac{n(n+1)}{2}\cdot(n^2+n+4)}{(2n+1)\cdot n\left[\frac{(n+1)(2n+1)}{6}+2\right]} \Rightarrow \lim_{n\to\infty} P(A) = \frac{3}{4}.$$

Q.9 (B)

$P(H) = \frac{3}{5}$; $P(T) = \frac{2}{5}$

E : Two consecutive heads for the first time

$P(E) = P\{(H\,H \text{ or } H\,T\,H\,H \text{ or } H\,T\,H\,T\,H\,H + \ldots\ldots) \text{ or } (T\,H\,H \text{ or } T\,H\,T\,H\,H \text{ or } \ldots\ldots)\}$

$$= \frac{P(H\,H)}{1-P(H\,T)} + \frac{P(T\,H\,H)}{1-P(H\,T)} = \frac{P(H)\cdot P(H)}{1-P(H)\,P(C)}\,[1+P(T)] = \frac{\frac{9}{25}}{1-\frac{6}{25}}\left(1+\frac{2}{5}\right) = \frac{9}{19}\cdot\frac{7}{5} = \frac{63}{95}$$ **Ans.**

Q.10 (A)

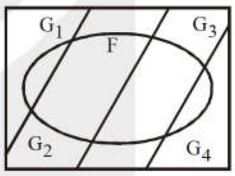
$P(A) = 0.2$; $P(B) = 0.4$; $P(C) = 0.6$

$P(H_1) = P(A\cap\overline{B}\cap\overline{C}) = (0.2)\,(0.6)\,(0.4)$

$P(H_1) = 0.048$

$P(H_2) = P(B\cap\overline{C}\cap\overline{A}) = (0.4)\,(0.8)\,(0.4) = 0.128$

$P(H_3) = P(C\cap\overline{A}\cap\overline{B}) = (0.6)\,(0.8)\,(0.6) = 0.288$



$$\therefore\ P\left(H_1/(H_1\cup H_2\cup H_3)\right) = \frac{0.048}{0.048+0.128+0.288} = \frac{48}{468} = \frac{3}{29}$$

$$P\left(H_2/(H_1\cup H_2\cup H_3)\right) = \frac{0.128}{0.468} = \frac{8}{29}$$

$$P\left(H_3/(H_1\cup H_2\cup H_3)\right) = \frac{18}{29}.$$ **Ans.**

Q.11 (B)

Q.12 (D)

A : sum on dice is 6, 12, 18

B : sum on dice is 12, 18

(12 and 18 has maximum number of divisors)

C : sum on two of the dice is 11 or more

$\therefore$ It is 12, 13, ... 18.

$$12 \Rightarrow \begin{cases} 1, 6, 5 & \rightarrow & (6 \text{ cases}) \\ 2, 6, 4 & \rightarrow & (6 \text{ cases}) \\ 2, 5, 5 & \rightarrow & (3 \text{ cases}) \\ 3, 6, 3 & \rightarrow & (3 \text{ cases}) \\ 3, 5, 4 & \rightarrow & (6 \text{ cases}) \\ 4, 4, 4 & \rightarrow & \underline{(1 \text{ cases})} \\ & & 25 \text{ cases} \end{cases} \qquad 18 \Rightarrow 6, 6, 6 \ (1 \text{ case})$$

$$6 \Rightarrow \begin{cases} 1, 4, 1 & \rightarrow & (3 \text{ cases}) \\ 1, 3, 2 & \rightarrow & (6 \text{ cases}) \\ 2, 2, 2 & \rightarrow & \underline{(1 \text{ cases})} \\ & & 10 \text{ cases} \end{cases}$$

(i) $$P\left(\frac{B}{A}\right) = \frac{P(\text{sum 12 or 18})}{P(\text{sum 6 or 12 or 18})} = \frac{25+1}{25+1+10} = \frac{26}{36}$$

(ii) $$P\left(\frac{B \cap C}{A}\right) = \frac{P(B \cap C \cap A)}{P(A)}$$

$$= \frac{P(\text{sum 12 or 18 when two of the dice have sum 11 or more})}{P(6 \text{ or } 12 \text{ or } 18)} = \frac{7}{36}$$

$12 \Rightarrow 5, 6, 1 \rightarrow (6 \text{ cases})$

$18 \Rightarrow 6, 6, 6 \rightarrow (1 \text{ cases})$.

Q.13 (C)

Q.14 (A)

Q.15 (B)

$P(A) = P(3, 6) = \frac{1}{3}$ **[Special Test-2]**

$P(B) = P(\text{Red and Blue die have distinct scores}) = 1 \cdot \frac{5}{6} = \frac{5}{6}$

$P(C) = P(\text{sum of scores on 3 dice is 10})$

i.e. $x + y + z = 10$; $n(S) = 216$

$n(C) = \text{coefficient of } x^{10} \text{ in } (x + x^2 + x^3 + x^4 + x^5 + x^6)^3$

$= \text{coefficient of } x^7 \text{ in } (1 + x + x^2 + x^3 + x^4 + x^5)^3$

631	→ 6
622	→ 3
541	→ 6
532	→ 6
433	→ 3
442	→ 3
Total = 27	

$$P(C) = \frac{27}{216} = \frac{1}{8}$$

Now $P(A/C) = \frac{P(A \cap C)}{P(C)}$

$A \cap C = \{631, 613, 361, 316, 622, 352, 325, 334, 343\}$

$P(A \cap C) = \frac{9}{216} = \frac{1}{24}$; $\qquad P(A/C) = \frac{1}{24} \cdot \frac{8}{1} = \frac{1}{3}$.

Q.16 (ABC)

$$\left\{\frac{6^p-1}{5}\right\}=0 \Rightarrow p=1,2,3,4,5,6$$

$$(p-5)(p^2-4p+3)=0 \Rightarrow p=1,3,5$$

$$\left[\frac{p-2}{3}\right]=0 \Rightarrow 2\le p<5 \Rightarrow p=2,3,4.$$

Q.17 (ABCD)

$$P(A)=\frac{1}{2};\ P(B)=\frac{1}{5};\ P(A\cap B)=\frac{1}{10};\ P(A\cup B)=\frac{3}{5}$$

$$\text{(A) } P\left(\frac{A}{A\cup B}\right)=\frac{P(A\cap(A\cup B))}{P(A\cup B)}=\frac{P(A)}{P(A\cup B)}=\frac{5}{6}$$

$$\text{(B) } \left(\frac{A\cap B}{A'\cup B'}\right)=\frac{P[(A\cap B)\cap(A\cap B)']}{P(A'\cup B')}=0$$

$$\text{(C) } P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=\frac{1}{2}$$

$$\text{(D) } P(A'\cap B')=1-P(A\cup B)=\frac{2}{5}$$

Q.18 (ACD)

4 defective + 11 correct

$$\text{P(getting existally three defective in 8 record)}=\frac{{}^4C_3\ {}^{11}C_5}{{}^{15}C_8}$$

$$\text{P(9}^{\text{th}}\text{ examined is last defective)}=\frac{{}^4C_3\ {}^{11}C_5}{{}^{15}C_8}\times\frac{1}{7}$$

Q.19 [1]

P(event A occurs) = 0.2; n= 3

Let E_1 : event A occur exactly once

$P(B/E_1)=0.1$ ....(1)

E_2 : event A occurs twice

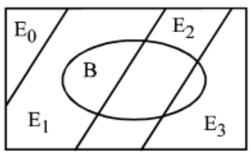
$P(B/E_2)=0.3$ ....(2)

E_3 : event A occurs three times

$P(B/E_3)=0.7$ ....(3)

E_0 : event A does not occurs

$P(B/E_0)=0$



$P(B)=P(B\cap E_1)+P(B\cap E_2)+P(B\cap E_3)$

$P(B)=P(E_1)\cdot P(B/E_1)+P(E_2)\cdot P(B/E_2)+P(E_3)\cdot P(B/E_3)$ ......(4)

Now $P(E_1)={}^3C_1\,(0.2)\,(0.8)^2$ $[\,{}^nC_r\,p^r\,q^{n-r}\,]$

$=(3)\,(0.2)\,(0.64)=0.384$

$P(E_2)={}^3C_2\,(0.2)^2\,(0.8)=(3)(0.04)(0.8)=0.096$

$P(E_3) = {}^3C_3\,(0.2)^3 = 0.002$

$\therefore \quad P(B) = (0.384)(0.1) + (0.096)(0.3) + (0.002)(0.7)$

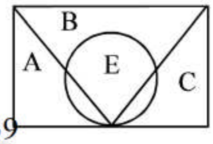
$= 0.0384 + 0.0288 + 0.0014 = 0.0686$

Hence $P(B \cap E_1) = 0.0384$ which has the maximum value.

Hence most probable number of occurrences of A is 1. **Ans.**

Q.20 [17]

Let A, B, C be the events that the radio set chosen, is manufactured by the companies A, B and C respectively. Let E be the event that the radio set is of excellent quality.



Thus $P(A) = \dfrac{18}{50}$; $P(B) = \dfrac{20}{50}$; $P(C) = \dfrac{12}{50}$

$P(E/A) = 0.9$; $P(E/B) = 0.6$; $P(E/C) = 0.9$

$\therefore \quad P(E) = P(A)\,P(E/A) + P(B)\,P(E/B) + P(C)\,P(E/C) = \dfrac{39}{50}$

required probability $= P(B/E) = \dfrac{P(B)\,P(E/B)}{P(E)} = \dfrac{4}{13}$ Ans.